

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 14B.3

BUFFER SOLUTIONS AND PH CURVES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	14: Acid-Base Equilibria
SUB-TOPIC:	14B.3: Buffer Solutions and pH Curves
TOTAL QUESTIONS:	1 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	1 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 14B.3

Q1. State the significance of the half-equivalence point in a weak acid – strong base titration.

[1 Marks]

WCH14/01/Oct21/Q19(e)

SECTION B: WORKED MARK SCHEME — 14B.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: State the significance of the half-equivalence point in a weak acid – strong base titration. At the half-equivalence point, $\text{pH} = \text{pK}_a$.	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 14B.3**FAQ #1: Where is the buffer region on a titration curve?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Where is the buffer region on a titration curve?

MODEL ANSWER / EXAMINER EXPECTATION:

It is the relatively flat section before the equivalence point where the weak acid is in excess and acting as a buffer.